

# MIMIC-III ICU Mortality Prediction Project

Pre-modeling data engineering, feature engineering and data quality control process

**First 24 hours**

**45,253 ICU stays**

**330M+ records**

Goal: build a reliable ML dataset for in-hospital mortality prediction using the first 24 hours of clinical data after ICU admission.

## 1. Project goal and study cohort

Defining the right patient population and time window before modeling

**45.253**

Adult ICU stays

$\geq 24$  hours  
Minimum ICU  
length of stay

First 24 hours  
Feature time  
window

11.84%  
In-hospital  
mortality

### Cohort logic

Instead of using all MIMIC-III records, adult stays with at least 24 hours in the ICU were selected.

The main cohort was fixed as the project's official ICU list in a separate Parquet file.

All feature tables were limited to these 45,253 ICU stays. To avoid using information close to or after the outcome, the data window was restricted to the first 24 hours after ICU admission.

### Why was the main cohort needed?

To distinguish true exclusions from missing data when patients disappeared from lab or vital tables  
To keep all sources within the same ICU universe  
To prevent patient loss during the final merge

## 2. ID structure and the multiple ICU stay problem

In MIMIC-III, “the same patient” is not represented by a single ID



**SUBJECT\_ID**  
Patient identifier



**HADM\_ID**  
Hospital admission



**ICUSTAY\_ID**  
ICU stay

**2.399**  
Multiple ICU stays  
within one  
HADM\_ID

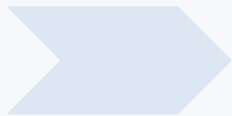
### ISSUE

Joining only by HADM\_ID could incorrectly assign measurements across multiple ICU stays within the same hospital admission.

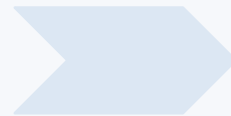
### FIX

A time filter was added to ID matching: a measurement was linked to an ICUSTAY\_ID only if it fell between that ICU INTIME and the first 24 hours.

**SUBJECT\_ID**



**HADM\_ID**

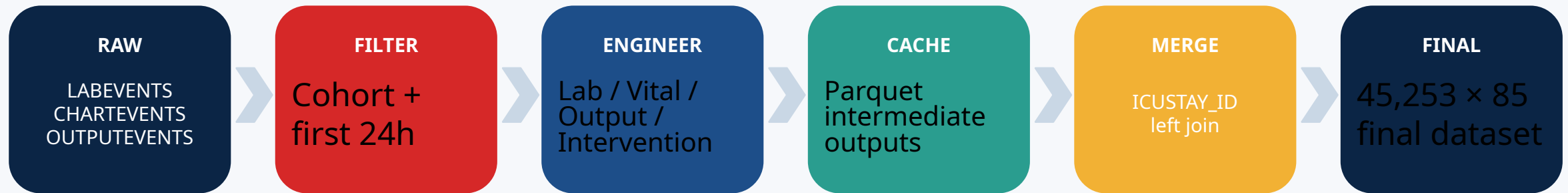


**ICUSTAY\_ID + INTIME**

Result: hospital-admission level data was safely mapped to the ICU-stay level.

### 3. Modular clinical data pipeline

From raw MIMIC tables to independent, reusable feature tables



**ISSUE** Because the raw CSVs were very large, reprocessing them from scratch in every notebook was slow and risky for memory.

**FIX** Typed Parquet intermediate outputs were produced after each main stage; later stages used these cleaned tables directly.

#### Main saved feature tables

- lab\_features\_first24h.parquet
- vital\_features\_first24h.parquet
- output\_features\_first24h.parquet
- intervention\_features\_first24h.parquet
- demographic\_features.parquet

This structure made it possible to rebuild the final dataset without returning to the raw tables.

## 4. Lab feature selection

Building a clinically and statistically meaningful core set from hundreds of tests



### Selection logic

Coverage was measured by how many distinct HADM\_IDs contained each test, not by row count.

Common tests were reviewed one by one for clinical importance and redundancy.

Repeated signals such as hemoglobin-hematocrit/RBC, INR-PT/PTT, and glucose variants were simplified.

AST, ALT, bilirubin and albumin were removed due to high missingness; lactate and blood gases were retained because of clinical importance.

### Final core tests

Hb • WBC • Platelet • Creatinine • BUN •  
Na/K/Cl • Bicarbonate • Anion Gap • Glucose  
• Mg • Ca • Phosphate • INR • pH • pO<sub>2</sub> • pCO<sub>2</sub>  
• Lactate

## 5. Lab data engineering and feature generation

Linking nearly 10 million selected lab records to the correct ICU stay and time window

**9.978.641**  
Selected lab  
records

**1.735.994**  
First 24h records

**500K**  
Chunk size

**44.626 × 39**  
Final lab table

**ISSUE** Merging ~10M rows with the ICU map at once caused WSL/RAM crashes.

**FIX** LABEVENTS was processed in 500K-row chunks; each chunk was matched by HADM\_ID and filtered by the CHARTTIME, INTIME + 24h window.

Feature design

First  
measurement  
**first**



Change  
**last – first**

Negative change values are not errors; they show clinical direction.

## 6. Vital signs: working with 330M+ records

Memory-friendly streaming extraction for data beyond pandas limits

**330.712.483**  
CHARTEVENTS  
record count

**7.031.935**  
Selected first 24h  
vital records

**44.549 × 33**  
Final vital table

**8 vital**  
Canonical feature  
groups

### ISSUE

Attempts to read with pandas in 500K chunks produced parsing/memory errors after 5-6 minutes.

### FIX

Strict PyArrow streaming scanned the file in small blocks; only selected ITEMIDs were kept and a Parquet cache was produced.

## CareVue + MetaVision

ITEMIDs for the same clinical measurement across systems were combined under one canonical feature.

Heart Rate, Respiratory Rate, SpO2, non-invasive SBP/DBP/MBP, Temperature and GCS Total were selected.

Temperature C/F units were normalized.

Vital features were generated with first / last / min / max summaries.

## 7. Output features: urine, chest tube and EBL

Turning output data with different recording structures into clinically meaningful features

### Urine output

Foley, Void, OR urine and other appropriate urine ITEMIDs were combined.  
The first 24h total was calculated at ICUSTAY\_ID level.  
Records with NaN values being treated like "0" were corrected, preserving the true missing count as 2,750.  
A single -1500 mL record was removed: the related stay total changed from -300 to 1,200 mL.  
A 555,975 mL gross error was removed: 561,190 to 5,215 mL.

### Chest tube

Amount data was missing for most of the cohort.  
Because of 82%+ missingness, a presence/absence approach was chosen instead of imputing total amount.  
8,106 ICU stays had chest\_tube\_present = 1.

### Decision

A binary feature carrying record presence was considered more reliable than weak amount information.

### Estimated Blood Loss (EBL)

ebl\_present and ebl\_24h were kept together.  
4,291 ICU stays had EBL records.  
Median  $\approx$  300 mL; max 50,000 mL.  
p99: 10,000 • p99.5: 13,685 • p99.9: 25,000  
If ebl\_present=0, ebl\_24h=0 in the final table.  
Extreme values were not removed; they were flagged for the modeling stage.

## 8. Intervention and demographic features

Adding high-signal clinical variables without expanding the scope too far

### Intervention features



CHARTEVENTS + INPUTEVENTS\_CV/MV + PROCEDUREEVENTS\_MV + OUTPUTEVENTS were used together. CareVue and MetaVision sources were combined under one binary feature.

### Demographics + admission

age  
gender  
admission\_type  
admission\_location  
Intentionally excluded  
insurance • marital status • ethnicity • religion • language

### Age policy

INTIME - DOB; due to MIMIC anonymization, ages 90+ were represented as 90 (2,239 stays).

## 9. Missingness: preserving meaning instead of deleting

Missingness can sometimes carry clinical signal in ICU data

Lab / Vital coverage  
intersection

| Group                       | Stay   | Mortality |
|-----------------------------|--------|-----------|
| Lab present / Vital present | 44,187 | %11,8     |
| Lab present / Vital missing | 439    | %12,3     |
| Lab missing / Vital present | 362    | %22,7     |
| Both missing                | 265    | %1,9      |

### Decision

Patients were not removed because of missingness.

Two group-level indicators were added: lab\_missing and vital\_missing.

Individual lab/vital NaNs were preserved in the final table.

Median imputation was left to the model preprocessing stage.

EBL was not treated as standard missingness; if ebl\_present=0, ebl\_24h=0.

Why? The 22.7% mortality in the lab missing / vital present group showed that missingness was not completely random.

## 10. Data quality audit: separating “outliers” from “errors”

Issues found after the final merge were fixed upstream at the source stage

**ISSUE** Impossible values were found, including negative Celsius temperatures, HR=5409, RR=2,355,555, SpO2=981,023 and BP values above 120,000.

**FIX** Native-unit vital limits were applied before conversion/aggregation; measurements marked as ERROR were set to NaN. 6,840 vital feature cells changed.

**ISSUE** The lab data contained clinically invalid records such as 0 hemoglobin and 0 INR.

**FIX** Only clearly invalid  $\leq 0$  Hb/INR values were set to NaN in source lab records; 6 feature cells were updated.

**ISSUE** Single gross-error urine records of -1500 mL and 555,975 mL were found.

**FIX** Negative/gross-error records were filtered upstream in the output pipeline; Parquet outputs were regenerated.

Values intentionally retained

Negative \*\_change • negative anion gap • extreme but clinically plausible lab/vital values • EBL max 50,000 mL • some very high urine totals

## 11. Leakage and final validation

### Preventing the model from “seeing the future”

Outcome information excluded from predictors

DOD / DOD\_HOSP / DOD\_SSN  
DEATHTIME  
DISCHTIME  
DISCHARGE\_LOCATION  
EXPIRE\_FLAG-type outcome fields

Target only: HOSPITAL\_EXPIRE\_FLAG

Final validation

45,253 unique ICUSTAY\_IDs  
Duplicate ICU = 0  
Missing target = 0  
Infinite values = 0  
No suspicious predictors other than the target in leakage screening  
Source tables regenerated after source-level cleaning

## 12. Final dataset and transition to modeling

Data engineering / feature engineering stage completed

**45.253**

Rows / ICU stays

**85**

Total columns

**83**

Predictors

**5.360**

Hospital deaths

### Final data structure

Lab + vital + output + intervention +  
demographics in one table  
Left join on ICUSTAY\_ID  
lab\_missing / vital\_missing retained  
EBL missingness handled separately  
424,942 true clinical missing cells preserved

### Next step: modeling

Median imputation (learned preprocessing)  
Categorical encoding  
Patient-level train/validation/test split  
Comparison of multiple models  
Evaluation of class imbalance and outlier  
strategy during modeling

# What does this project demonstrate?

## **BIG DATA**

Memory-friendly streaming and chunk processing over 330M+ records

## **CLINICAL FEATURE ENGINEERING**

Meaningful selection from hundreds of tests, time-window and trajectory features

## **DATA QUALITY**

Resolving gross errors, unit issues, missingness and outlier distinctions at the source

## **LEAKAGE PREVENTION**

Excluding post-outcome information and preserving the first 24h window

## **MODULAR PIPELINE**

Reusable and auditable process with Parquet intermediate tables

## **PROBLEM SOLVING**

Solving memory crashes, parsing errors, multi-stay matching and data inconsistencies

**Result: before modeling, a reliable, traceable and clinically meaningful ML dataset was created.**